

Airmar RS-232 Altimeter

Some Slocum G3S Gliders are now being built with the Airmar RS-232 Altimeter board – ‘altimeter_232’ (TWR part number: 313454-NFC). The externally mounted altimeter transducer remains the same, but the difference is that the traditional altimeter board in the forward section of the glider (Airmar Analog AD Altimeter board (TWR part number: E-388)) has been replaced with a newer, digital board. This document is meant to highlight major differences in the new board.

Hardware Differences:

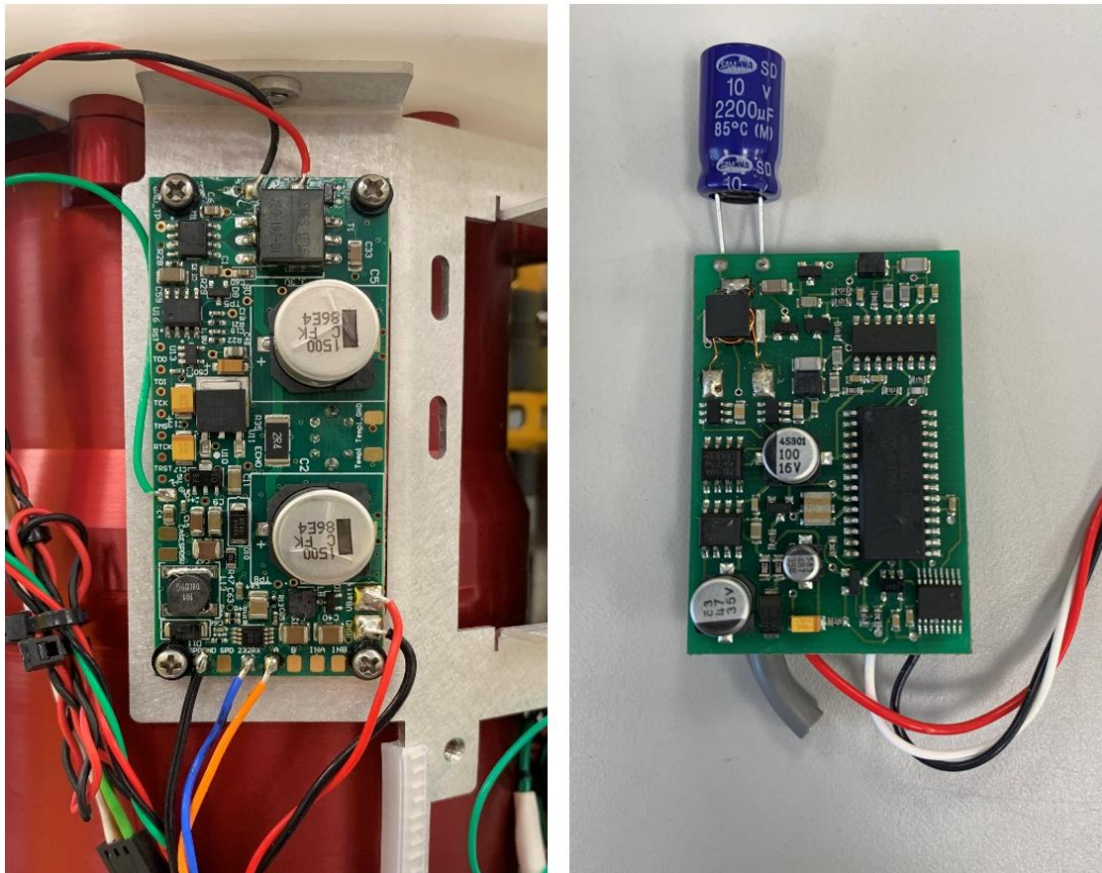


Figure 1. The new Airmar RS-232 Altimeter Board connected to forward pump (left). The traditional Airmar Analog AD Altimeter Board (right).

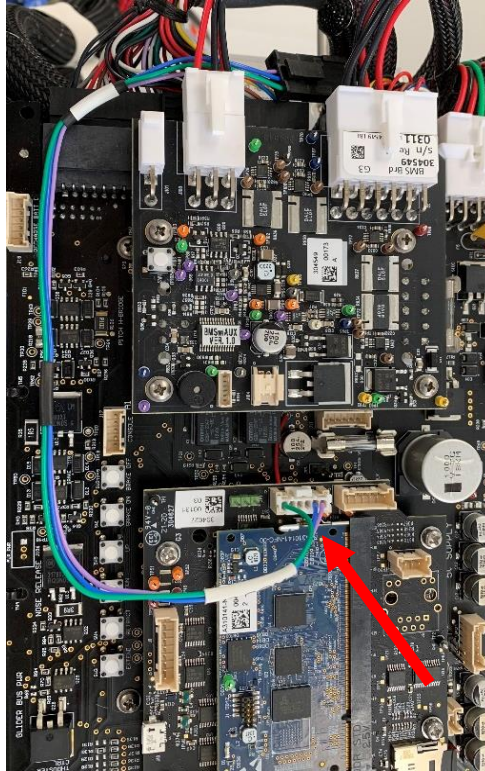


Figure 2. The '305912-NFC' connector, between the aft section and science bay, has been modified to support and additional altimeter connector which plugs into SOM carrier board. This configuration is also compatible with the traditional Airmar Analog AD Altimeter Board

Software Differences:

- The glider needs to have Software Release V10.07 (or later) installed.
- 'altimeter_232' needs to be listed under installed devices in autoexec.mi and 'altimeter' should be removed.
- Variables in autoexec.mi – TWR recommends that the following lines be added to autoexec.mi to adjust sensor values from the masterdata default when using the altimeter_232:

```
sensor: f_airmar_altimeter_time_until_good_reading(s)    12           # to adjust for the longer boot up
                                                                # time of the altimeter_232

sensor: u_max_altimeter(m)          200                         # to account for the larger expected range
                                                                # on the altimeter_232
```

- Example of a device menu with altimeter_232 installed:

```

use
    name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
    [I Installed] [- Not_Installed]
    [u In_use] [- Not_In_use] [X Out_of_Service]
    name      limits      stats (#total/#mission/#segment)
              os w/s w/m os      errors      warnings      oddities
0      simdrv -
1      test_driver -
2      ARGOS* I u  -1  20   5  0
3      WATCHDOG I u  -1  -1  -1  0
4      DEADMAN I u  -1  20   5  0
5      CONSOLE* I u  -1  20   5  0
6      GPS I u  -1  20   5  0
7      attitude_rev I u  3  20   5  0
8      ocean_pressure I u  3  20   5  0
9      vacuum I u  3  20   5  0
10     battery I u  3  20   5  0
11     air_pump I u  3  20   5  0
12     pitch_motor I u  3  20   5  0
13     science_super I u  3  20   5  0
14     digifin I u  3  20   5  0
15     altimeter -
16     altimeter_232 I u  3  20   5  0
17     IRIDIUM* I u  -1  20   5  0
18     leakdetect I u  3  20   5  0
19     recovery -
20     coulomb I u  3  20   5  0
21     veh_temp I u  3  20   5  0
22     BUOYANCY_PUMP -

23     DE_PUMP -
24     HD_PUMP I u  3  20   5  0
25     thruster I u  3  20   5  0
devices:(t/m/s) errs:  0/  0/  0 warn:  0/  0/  0 odd:  0/  0/  0

```

Operational differences:

- In a mission, the 'altimeter_232' should behave the same as the Analog AD 'altimeter'.

```

c_alt_time = 0      # altimeter should be sampling as fast as possible
c_alt_time = -1     # altimeter should be off

```
- We still encourage `u_alt_reduced_usage_mode = 1` in most scenarios, so that the altimeter is not powered on more than needed.
- `m_altimeter_voltage` will not produce output.
- 'adtest altimeter' is obsolete. 'talk altimeter' is supported.
- The cadence of the altimeter_232 clicks is slightly different when listening to its operation.
- The altimeter_232 is expected to have further range than the traditional altimeter. Teledyne Webb Research has operationally observed a range up to 130m range in testing, though greater ranges may be possible.

- Useful variables related to the altimeter (see masterdata for descriptions):
 - `c_alt_time`
 - `m_altimeter_status`
 - `m_raw_altitude`
 - `m_altitude`
 - `m_water_depth`
 - `u_alt_min_depth`
 - `u_alt_reduced_usage_mode`
 - `u_alt_debug`
- A reminder that the masterdata default for `c_alt_time` is currently 0 (it defaults to being 'on' when the glider boots up) – it is suggested to 'put `c_alt_time -1`' to power the altimeter board off if the glider will be powered on and sitting in air for a while.
- This will not be an issue if `c_alt_time` is being set by a mission, but if the user is turning the altimeter ON by setting `c_alt_time ≥ 0`, the following will apply: The altimeter_232 takes slightly longer to boot up than the traditional board. As a result, `c_alt_time` must be greater than `f_airmar_altimeter_time_until_good_reading` to allow the altimeter to power down between samples. For example, if `c_alt_time = 10` seconds and `f_airmar_altimeter_time_until_good_reading = 12` seconds, the altimeter will not power down between samples. If `c_alt_time = 15` seconds and `f_airmar_altimeter_time_until_good_reading = 12` seconds, the altimeter will power down between samples.

Functional Testing on altimeter_232:

On the analog AD altimeter – reporting voltage and listening for clicking was a good way to verify the health of the altimeter board and transducer. On the altimeter_232, you will not see 'm_altimeter_voltage' updating. The following is the recommended procedure for functionally testing the altimeter_232:

- `put c_alt_time 0      # power on the altimeter`
- `put u_alt_debug 1      # put altimeter debug into verbose mode`
- Verify that altimeter transducer is clicking.
- Verify altimeter debug is outputting 'altimeter_232_measure, received '\$SDDPT.....', and 'altimeter_232_measure() RX valid message '\$SDDPT.....' lines.
- `put c_alt_time -1      # power off altimeter`
- Wait to see 'altimeter_232_power_off(), power removed, uart closed' from debug output.
- Verify altimeter has stopped clicking
- `put u_alt_debug 0      # disable verbose altimeter debug`

Glider Modularity with the altimeter_232:

- To use a forward section with an altimeter_232 with an aft section not shipped configured for the altimeter_232, a modified '305912-NFC' connector (Rev.C) needs to be installed on the aft section. The appropriate software and setting updates will also need to be made, see the 'Software Differences' section of this document.
- To use an aft section configured for an altimeter_232 with a forward section with the traditional Airmar Analog AD Altimeter, the modified '305912-NFC' connector (Rev. C) is backwards compatible and can be used with the Airmar Analog AD Altimeter. The appropriate software and setting adjustments will also need to be made, see the 'Software Differences' section of this document to reverse – this includes removing 'altimeter_232' and replacing it with 'altimeter' under installed devices in autoexec.mi. Additionally, masterdata default values should be used for f_airmar_altimeter_time_until_good_reading (default = 4 s) and u_max_altimeter (default = 100 m), this can be achieved by commenting out these sensors in the autoexec.mi file . Software Release V10.07 and later will still support the Airmar Analog AD Altimeter.

Appendix:

Values for M_ALTIMETER_STATUS

```
#define AS_GOOD_VALUE 0.0 // Altimeter reading is good,
#define AS_BPUMP_MOVING 1.0 // The buoyancy pump is moving
#define AS_NOT_DIVING_OR_HOVERING 2.0 // We aren't diving or hovering
#define AS_INFLECTING 3.0 // We are inflecting now
#define AS_POST_INFLECTION 4.0 // inflected in U_ALT_MIN_POST_INFLECTION_TIME secs
#define AS_TOO_SHALLOW 5.0 // vehicle depth is above U_ALT_MIN_DEPTH
#define AS_SHORT_READING 6.0 // the altimeter reads < U_MIN_ALTIMETER
#define AS_LONG_READING 7.0 // the altimeter reads > U_MAX_ALTIMETER
#define AS_WD_TOO_SHALLOW 8.0 // the altitude results in water depth < U_MIN_WATER_DEPTH
#define AS_WD_TOO_DEEP 9.0 // the altitude results in water depth > U_MAX_WATER_DEPTH
#define AS_BOT_CHANGED_TOO_FAST 10.0 /* altitude results in a change in water depth that
is greater than u_max_bottom_slope times the presumed horizontal movement of the glider
(using nominal speed)
#define AS_WD_UNKNOWN_REASON 11.0 // Something to do with water depth, but don't know
what
#define AS_NOT_ENUF_IN_ROW 12.0 // haven't had U_ALT_REQD_GOOD_IN_A_ROW good readings
like 1
```

Example of debug output with altimeter off (c_alt_time -1)

```
48.70    altimeter_232_ctrl()  
52.70 10 altimeter_232_ctrl()  
56.70 11 altimeter_232_ctrl()  
60.71 12 altimeter_232_ctrl()  
64.71 13 altimeter_232_ctrl()  
68.71 14 altimeter_232_ctrl()
```

Example of debug output with altimeter on in air (c_alt_time 0)

```
966.04    altimeter_232_ctrl()  
966.04    altimeter_232_measure()  
966.40    altimeter_232_ctrl()  
966.40    altimeter_232_measure()  
966.40    altimeter_232_measure, received '$SDDPT,,, *7B'  
966.40    altimeter_232_measure() RX valid message '$SDDPT,,, *7B'  
966.41    altimeter_232_power_off()  
968.04 35 altimeter_232_ctrl()
```